

Eco-Series-Kitty — Version Control & Release Guide

Repository:https://github.com/Rahul-ambiplatforms/Eco-Series-Kitty

Product:Kitty Camera Firmware (Augentix ARM)

CI/CD:GitHub Actions (release.yml)

Last Updated:March 2026

1. Repository Structure

```
Eco-Series-Kitty/
├── mqtt-client/           # C source code (edit here)
│   ├── HTTP.c            # Main MQTT client logic (2300+ lines, 78 cases)
│   ├── N_HTTPS.c         # HTTP/HTTPS request handler
│   ├── N_HTTPS.h         # Header file
│   ├── Makefile          # Cross-compilation Makefile
│   └── deps/             # Build dependencies (headers + libs)
├── kitty/Augentix/AugenTix/ # Deployable package (auto-assembled by CI)
│   ├── MQTT_vcamclient_Augentix # Built binary (from mqtt-client/)
│   ├── P2Pambicam          # P2P tunneling binary
│   ├── ambicam.sh          # Startup script
│   ├── config.json         # Version + MQTT config ← VERSION IS HERE
│   ├── lib*.so.*           # Shared libraries
│   └── *.crt, *.key        # TLS certificates
├── mqtt-server/          # Node.js MQTT broker (server-side)
├── toolchain/            # ARM cross-compiler (GCC 10.3.0)
├── bin/                  # Built binaries
├── docs/                 # Documentation
└── .github/workflows/
    └── release.yml        # CI/CD pipeline
```

2. Version Format

Version format: A_Series_MAJOR.MINOR.PATCH

Example: A_Series_0.10.7 → Major=0, Minor=10, Patch=7

Component	When it changes	Example
MAJOR	Breaking changes (hardware change, protocol change)	0.10.7 → 1.0.0
MINOR	New feature added	0.10.7 → 0.11.0

PATCH	Bug fix, small improvement, config change	0.10.7 → 0.10.8
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3. How the CI/CD Works



3.1 Auto Version Bump Rules

Your commit message contains	Bump type	Version change
feat or feature	FEATURE (minor)	0.10.7 → 0.11.0
fix, bug, or hotfix	BUG (patch)	0.10.7 → 0.10.8
anything else	PATCH (patch)	0.10.7 → 0.10.8

4. How to Push, Commit & Release

Example 1: Bug Fix (Patch Release)

Scenario: You fixed a buffer overflow in HTTP.c

Result: Version 0.10.7 → 0.10.8

```
# 1. Edit the code
nano mqtt-client/HTTP.c

# 2. Stage your changes
git add mqtt-client/HTTP.c
```

```
# 3. Commit with "fix" in the message
git commit -m "fix: resolve buffer overflow in write_callback"

# 4. Push to main
git push

# 5. Done! CI automatically:
#   → Detects "fix" in commit message
#   → Bumps 0.10.7 → 0.10.8
#   → Builds binary
#   → Creates Augentix.tar.gz
#   → Tags v0.10.8
#   → Publishes GitHub Release
```

Example 2: New Feature (Minor Release)

Scenario: You added a new MQTT case 79 for audio config

Result: Version 0.10.8 → 0.11.0

```
# 1. Edit the code
nano mqtt-client/HTTP.c

# 2. Stage your changes
git add mqtt-client/HTTP.c

# 3. Commit with "feat" in the message
git commit -m "feat: add case 79 for audio volume control"

# 4. Push
git push

# 5. CI bumps minor: 0.10.8 → 0.11.0
```

Example 3: Update ambicam.sh (Patch Release)

Scenario: You updated the startup script

Result: Version 0.11.0 → 0.11.1

```
# 1. Edit the startup script
nano kitty/Auentix/AugenTix/ambicam.sh

# 2. Commit and push
git add kitty/Auentix/AugenTix/ambicam.sh
git commit -m "update: increase P2P retry timeout to 10 attempts"
git push

# 3. CI bumps patch: 0.11.0 → 0.11.1
```

Example 4: Manual Release (from GitHub UI)

Scenario: You want to release without code changes, or re-release

```
# 1. Go to GitHub → Actions tab
# 2. Click "Build & Release Kitty Firmware"
# 3. Click "Run workflow"
# 4. Select bump type: patch / feature / bug
# 5. Click "Run workflow" button
```

5. Commit Message Cheat Sheet

What you did	Commit message	Version bump
Fixed a crash	fix: null pointer crash in onConnectionLost	Patch (+0.0.1)
Fixed a memory leak	bugfix: free cJSON memory in get_system_info	Patch (+0.0.1)
Added new MQTT case	feat: add case 79 for speaker volume	Minor (+0.1.0)
Added PTZ preset feature	feature: support PTZ preset tour	Minor (+0.1.0)
Changed PTZ speed	update PTZ speed from 6 to 5	Patch (+0.0.1)
Updated startup script	update ambicam.sh retry logic	Patch (+0.0.1)
Updated library	update libcurl to 8.7.0	Patch (+0.0.1)
Security hotfix	hotfix: validate OTA URL to prevent injection	Patch (+0.0.1)

Important: The keyword must appear anywhere in the commit message (case-insensitive).
"fix: something", "Fixed the bug", "This is a bugfix" — all trigger patch bump.
"feat: new thing", "Added feature for PTZ" — all trigger minor bump.

6. Where to Find Releases

What	Where
All releases	github.com/.../releases
Download Augentix.tar.gz	Click release → Assets → Augentix.tar.gz
CI build logs	github.com/.../actions
Current version	kitty/Augentix/AugenTix/config.json → firmware field

7. Deploying to Camera

```
# 1. Download Augentix.tar.gz from GitHub Release

# 2. Upload to your OTA server
scp Augentix.tar.gz user@ota-server:/path/to/firmware/

# 3. Send OTA command to camera via MQTT (case 36)
mosquitto_pub -h prong.arcisai.io -p 1883 \
  -u Torque -P 'Raptor@0' \
  -t "torque/rx/CAMERA-SN/36" \
  -m "https://ota-server.com/firmware/Augentix.tar.gz"

# 4. Camera downloads, extracts, replaces files, reboots
```

8. Quick Reference

Daily workflow in 4 lines:

```
# Edit → Stage → Commit → Push
git add mqtt-client/HTTP.c
git commit -m "fix: resolve crash on large HTTP response"
git push
# Done. CI builds, packages, versions, and releases automatically.
```

Check current version:

```
cat kitty/Augentix/AugenTix/config.json | grep firmware
"firmware":"A_Series_0.10.7"
```

Check release history:

```
git tag --list 'v*' --sort=-version:refname
v0.11.0
v0.10.8
v0.10.7
```

9. Troubleshooting

Problem	Solution
CI build failed	Check Actions tab → click failed run → read error log
Release not created	Check if push was to main branch and changed mqtt-client/ or kitty/ files
Wrong version bumped	Use correct keyword in commit message: fix: for patch, feat: for minor

Want to re-release same version	Use manual trigger from Actions tab → "Run workflow"
Changes to mqtt-server/ don't trigger CI	Correct — only mqtt-client/ and kitty/ changes trigger builds
Version in config.json is wrong	Don't edit manually — CI auto-bumps it. If needed, edit and commit with <code>[skip ci]</code> in message

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